

PRENTISS, AB, Canada

WMO: 713530

Lat: 52.4336N

Lon: 113.5921W

Elev: 3166

StdP: 13.09

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-17.9	-12.6	-24.5	1.6	-17.6	-18.9	2.2	-12.3	25.3	26.9	21.8	23.8	7.5	280	0.679

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.2	80.8	62.5	77.5	60.7	74.7	59.4	65.7	76.7	63.4	74.2	61.3	71.5	7.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.4	91.5	72.4	58.9	83.6	69.2	56.7	77.1	67.1	32.3	77.1	30.5	74.5	28.8	71.7	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.3	20.1	17.7	DB	-23.4	86.9	5.1	3.5	-27.1	89.4	-30.0	91.4	-32.9	93.4	-36.5	95.9	(4)
(5)			WB	-23.7	70.0	5.0	2.1	-27.3	71.5	-30.3	72.7	-33.1	73.9	-36.7	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	38.8	17.8	16.0	27.2	38.9	50.6	57.0	61.6	59.3	51.7	40.2	26.3	17.9
	DBStd	19.60	15.23	14.62	12.52	9.57	8.13	5.44	5.28	5.95	8.46	9.43	12.85	13.57
	HDD50	5217	997	952	707	347	92	8	0	5	77	323	713	996
	HDD65	9604	1462	1372	1172	784	450	247	132	191	404	768	1161	1461
	CDD50	1144	0	0	1	13	110	216	360	294	128	20	2	0
	CDD65	56	0	0	0	0	3	6	27	15	5	0	0	0
	CDH74	772	0	0	0	2	66	71	296	227	105	5	0	0
	CDH80	119	0	0	0	0	6	5	51	37	20	0	0	0
Wind	WSAvg	8.8	9.4	8.8	8.9	9.9	9.9	8.8	7.5	7.2	8.1	8.9	9.0	9.1
Precipitation	PrecAvg	19.1	0.9	0.6	0.8	1.1	2.3	3.7	3.4	2.5	1.4	0.9	0.8	0.7
	PrecMax	23.3	2.5	1.6	2.0	2.9	4.7	7.0	7.5	4.6	3.1	2.5	2.4	1.7
	PrecMin	14.2	0.2	0.2	0.2	0.2	0.9	0.8	0.7	0.7	0.2	0.3	0.1	0.2
	PrecStd	2.4	0.5	0.3	0.4	0.6	0.9	1.3	1.5	0.9	0.8	0.5	0.5	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.6	47.8	56.4	71.0	80.0	79.2	84.9	83.8	82.9	72.4	59.1	47.0
		MCWB	37.3	38.2	43.9	49.6	54.2	61.2	66.8	62.3	58.3	52.0	45.8	38.0
	2%	DB	43.5	42.2	49.6	65.4	75.3	76.1	80.9	79.6	76.7	64.3	51.3	42.8
		MCWB	36.1	34.2	39.5	47.6	53.0	59.1	66.0	62.2	57.1	49.4	41.2	35.5
	5%	DB	41.0	38.6	45.6	59.2	70.8	72.5	77.1	76.1	71.7	58.8	45.3	39.6
		MCWB	34.4	32.1	37.4	43.7	51.4	57.4	63.7	61.4	55.4	46.4	37.1	33.1
	10%	DB	37.5	35.3	42.1	54.0	66.2	69.0	73.8	72.4	66.5	54.1	41.2	35.8
		MCWB	31.8	30.1	35.3	41.2	49.8	55.5	61.6	60.0	53.3	43.7	34.5	30.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.4	38.9	44.3	50.7	56.8	64.8	70.4	66.6	61.2	53.6	46.4	38.7
		MCDB	44.8	47.3	55.3	69.8	74.3	76.7	80.3	78.1	77.0	69.9	58.0	45.6
	2%	WB	36.5	34.9	40.3	47.8	54.1	61.3	67.2	64.5	58.2	49.9	41.5	35.8
		MCDB	42.8	41.2	48.3	64.2	71.5	72.1	78.1	76.0	73.0	63.5	50.1	42.4
	5%	WB	34.6	32.5	37.6	44.7	52.3	58.9	64.6	62.4	55.9	47.1	37.9	33.2
		MCDB	40.7	38.0	45.0	57.4	67.9	69.4	75.3	73.5	69.5	57.5	44.1	39.2
	10%	WB	32.0	30.2	35.5	42.0	50.5	56.8	62.4	60.2	53.8	44.4	35.2	30.4
		MCDB	37.3	35.2	41.7	52.5	64.1	66.7	72.4	71.3	65.3	53.1	40.4	35.5
Mean Daily Temperature Range	5% DB	MDBR	15.3	16.0	15.8	18.9	21.1	18.7	20.2	20.8	20.6	18.2	14.7	13.8
		MCDBR	16.1	18.4	20.2	28.1	28.4	24.0	24.8	27.3	29.1	25.8	19.9	15.4
		MCWBR	11.2	12.6	12.9	14.6	12.4	11.6	13.6	14.0	13.7	13.9	12.4	10.9
	5% WB	MCDBR	15.6	18.2	19.5	26.3	25.5	21.4	23.9	25.2	26.6	25.1	18.8	15.2
		MCWBR	11.3	13.0	12.9	14.0	11.5	11.6	13.9	14.1	13.2	13.9	12.5	11.0
Clear-Sky Solar Irradiance	taub	0.257	0.262	0.288	0.318	0.331	0.338	0.340	0.352	0.303	0.278	0.250	0.240	
	taud	2.321	2.397	2.425	2.391	2.410	2.413	2.434	2.374	2.491	2.544	2.477	2.328	
	Ebn at Noon	240	277	290	292	292	290	287	277	281	270	247	227	
	Edh at Noon	19	23	28	33	35	35	34	34	27	20	16	16	
All-Sky Solar Radiation	RadAvg	341	670	1115	1494	1814	1919	2013	1647	1145	684	350	251	
	RadStd	30	55	81	138	115	124	107	113	123	79	35	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (79 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page